



CSE 598 008 EECS 498 003 Lab 3

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Based on slides by Ivan Wei (Fall 2025).

Reminder

- Problem Set 1 due September 25
- Problem Set 2 will be released on September 26, due October 7
- Midterm will be on October 22, 6pm – 8pm, FMCRB 1050

Left-behind Content

- Safety Proof
- Jay Normal Form (JNF)

Jay Normal Form

Jay Normal Form

```
datatype Step =  
  | Action1Step( <parameters> )  
  | Action2Step( <parameters> )  
  ...  
  
ghost predicate NextStep(v: Variables, v': Variables, step:Step)  
{  
  match step  
  case Action1Step(<parameters>) => Action1(v, v', <parameters>)  
  case Action2Step(<parameters>) => Action2(v, v', <parameters>)  
  ...  
}  
predicate Next(v: Variables, v': Variables)  
{  
  exists step :: NextStep(v, v', step)  
}
```



VSCode transition

State Machine Exercise

- Starter code: `lab03a.dfy` and `lab03b.dfy` on the course GitHub

Extra - Sorites Paradox

- Base Case: Someone who is 2 m tall is a tall person.
- Inductive Step: Someone who is 1 mm shorter than a tall person is also tall
- Conclusion: Someone who is 1 m tall is a tall person.